

33A — Law Descent and Recoverability Novelty Audit

Professional reading edition of the evolving public MKUFT canon

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1. Audit question

The audit tests whether Modules 33S2 and 33S3 merely rename established results, or whether a residual operational structure survives after the strongest nearby prior art is granted in full.

The burden is deliberately asymmetric:

If prior work already supplies a component, that component is not claimed. Novelty belongs only to the residual conjunction that remains after subtraction.

The primary target is not vocabulary. It is the executable structure:

1. distinguish lineage, functional identity, constitutive closure, and dynamical law sufficiency;
2. test law sufficiency at a declared scale/address for a declared property and intervention class;
3. allow the active explanatory/dynamical address to move upward, downward, or multiscale when sufficiency changes;
4. keep local performance, whole-level closure, property-specific law sufficiency, and recoverability as separately testable objects;
5. treat recovery as target-relative constrained reachability, not assumed reversal;
6. require apparent irreducible history to survive ordinary state/memory augmentation before it is promoted.

2. Strong prior art that removes broad novelty claims

2.1 Micro-to-macro causal abstraction and intervention consistency are established

Rubenstein et al. (2017) formalise exact transformations between structural causal models at different levels and require consistency of interventional conclusions. Beckers and Halpern (2019), followed by Beckers, Eberhardt and Halpern (2020), develop exact, strong and approximate causal abstraction between low- and high-level models, explicitly including allowed interventions and micro-to-macro variable maps.

Boundary: 33S2 does not claim invention of micro/macro causal abstraction, intervention-consistent abstraction, exact transformations, approximate abstraction, or allowed-intervention maps.

2.2 Lumpability and dynamically valid coarse-graining are established

Markov-chain lumpability and related state-aggregation theory already ask when a projected/coarse state admits closed Markov dynamics. Strong and higher-order lumpability, approximate coarse-graining, aggregation error bounds, and de-aggregation are established topics.

Boundary: the 33S2 representative-consistency equations are not a new quotient or lumpability theorem. They instantiate a known mathematical burden inside the MKUFT addressed-law architecture.

2.3 Property- or goal-oriented reduction is established

Goal-oriented reduced-order modelling explicitly builds reduced descriptions to preserve a declared quantity of interest rather than every microscopic variable. Property-driven state-space coarsening similarly groups states according to preservation of declared trajectory properties.

Boundary: the raw statement “a coarse description may be sufficient for one property but not another” is not claimed as an unprecedented discovery.

This prior art materially narrows the novelty of **property-relative law descent in isolation**.

2.4 Macro causal efficacy and causal emergence are established

Hoel, Albantakis and Tononi (2013) show that macro causal models can outperform micro models under an effective-information measure despite supervening on the microdynamics.

Boundary: 33S2 does not claim invention of causal emergence, macro causal efficacy, or the possibility that a macro description can be scientifically superior for a declared purpose.

2.5 Memory repair and history-augmented reduced dynamics are established

Coarse-graining can create non-Markovian dynamics. Mori–Zwanzig methods explicitly recover unresolved influence through memory kernels, and higher-order lumpability/history-augmented descriptions are established.

Boundary: 33S2/33S3 do not claim invention of memory augmentation, non-Markovian coarse dynamics, higher-order Markov closure, or the idea that missing state can masquerade as path dependence.

2.6 Viability, reachability and recoverable/safe sets are established

Control and viability theory already distinguish admissible/safe state sets, viability, invariance, controllability and reachability under declared control classes and constraints.

Boundary: the binary recoverability scaffold Rec_H in 33S3 is not a new reachability theorem or a claim to have invented viability kernels.

2.7 Hysteresis, alternative stable states and recovery dynamics are established

Dynamical-systems and ecological work already treats hysteresis, alternative stable states, tipping transitions and changing recovery rates as distinct phenomena. Rindi et al. (2017), for example, directly measured increasing recovery length approaching ecological collapse.

Boundary: 33S3 does not claim invention of hysteresis, tipping points, recovery rates, alternative stable states, asymmetric return paths, or ecological resilience theory.

2.8 Multilevel fitness conflict and fitness decoupling are established

Multilevel-selection work explicitly demonstrates that collective-level fitness can become decoupled from constituent-cell performance. Hammerschmidt et al. (2014) experimentally found improved collective fitness without corresponding improvement in single-cell performance. Cancer and cell-competition literature likewise shows that cell-level competitive advantage and tissue-level integrity can align or conflict depending on context; normal epithelia can eliminate transformed cells, and mutant clones can even suppress emerging tumours.

Boundary: 33S3 does not claim invention of multilevel selection, fitness decoupling, cell competition, or the raw proposition that local success can conflict with whole-level success.

3. What survives for Module 33S2

3.1 The four-object separation survives as an operational conjunction

Prior literatures separately discuss historical identity, functional organisation/autonomy, closure, effective dynamics, coarse-graining and causal abstraction. In the literature examined here, no single source was located that supplies the 33S2 four-stage burden in the same operational form:

lineage → relational functional identity → constitutive closure → property-specific law sufficiency

The important residual is not that each distinction is individually unknown. It is that 33S2 prohibits one from licensing the next:

- common lineage does not establish preserved functional organisation;
- functional organisation does not establish constitutive/self-maintaining closure;
- closure does not establish autonomous higher-address dynamics for every property;
- predictive sufficiency does not establish interventional sufficiency.

Current novelty confidence: moderate candidate originality as an explicit integrated decision architecture; low-to-moderate novelty for the individual distinctions.

3.2 Property-relative law descent is narrowed but not eliminated

Property/goal-oriented model reduction is strong prior art. Therefore the claim that a reduced model may be adequate for one observable but not another is not novel by itself.

What remains distinctive in 33S2 is the use of the result as a **law-ownership gate** inside a recursively addressed system: the property test determines whether the promoted higher relational object may supply the active law object for that property, while lower substrate laws remain physically operative.

Current novelty confidence: low-to-moderate for property specificity alone; moderate for its integration with relational object promotion and addressed law assembly.

3.3 Predictive versus interventional descent has strong prior art

Causal-abstraction literature already places heavy emphasis on intervention consistency across levels.

Current novelty confidence: low in isolation.

Its role remains important because it prevents a passive predictive macro model from being promoted into causal/control authority merely because it forecasts well.

3.4 Bidirectional readdressing remains a candidate operational contribution

Adaptive multiscale modelling and coarse/fine model switching are established broad ideas, so “sometimes use a finer model” is not novel.

The narrower 33S2 rule is more specific: **the active address changes because the currently claimed higher-level law loses or gains demonstrated sufficiency for the declared property, preparation, intervention class, timescale or**

tolerance. A failed higher law does not imply that the higher functional object has ceased to exist; the model can move to a coupled multiscale description without ontological deletion.

No exact prior formulation of that full inference rule was located in this audit.

Current novelty confidence: moderate candidate operationalisation.

3.5 Closure margin versus law-descent margin is a strong residual discriminator

33S2 defines separate margins for:

- loss/fracture of the relation constituting the organised whole, M_C ;
- loss of higher-address law sufficiency for a declared property, $M_D(q)$.

The essential consequence is that a higher-order object can remain real and organised after the simpler macro law for a particular property has failed.

Adjacent fields contain pieces of this distinction, but the audit did not locate the same paired-margin object integrated with object promotion and readdressing.

Current novelty confidence: moderate candidate formal/operational contribution.

3.6 “Address before law ownership” is the strongest 33S2 residual

The literature searched contains no scientific use of the exact rule in the MKUFT sense:

construct/use the law at the coarsest relational address that has earned sufficiency for the property being propagated, and surrender that address when the sufficiency burden fails.

This should not presently be called a new physical law. It is better classified as a **candidate cross-scale formal principle / operational meta-rule for effective-law ownership**.

Current novelty confidence: moderate–high candidate synthesis relative to the literature examined; global priority remains open.

4. What survives for Module 33S3

4.1 Local gain versus whole-level gain is not novel in isolation

Multilevel selection and cancer biology already establish this strongly.

Current novelty confidence: low in isolation.

4.2 Recoverability as reachability is not novel in isolation

Viability and control theory already own this mathematical family.

Current novelty confidence: low in isolation.

4.3 Hysteresis and asymmetric return are not novel in isolation

Established dynamical-systems and ecological theory own these concepts.

Current novelty confidence: low in isolation.

4.4 The three-boundary separation survives as a useful integrated object

33S3 requires separate treatment of:

1. loss of property-specific higher-law sufficiency;
2. loss of constitutive closure;
3. loss of recoverability to a declared target under a declared intervention/environment/horizon class.

These events need not coincide and no universal ordering is assumed.

Each component has strong antecedents, but the audit did not locate this exact three-boundary construction embedded in a property-relative addressed-law architecture.

Current novelty confidence: moderate candidate integration.

4.5 Cross-scale alignment tuple survives as an operational bridge

The tuple

$(\Delta J_1, \Delta M_C, \Delta M_D(q))$

forces local performance, whole-level closure and higher-law sufficiency to be measured separately. The alignment classes — aligned, antagonistic, neutral/decoupled and mixed/property-relative — do not claim a universal biological or physical mechanism.

The novelty is not the existence of multilevel conflict; it is the integration of scale-typed performance with closure and law-descent margins in the same addressed update architecture.

Current novelty confidence: moderate candidate operational synthesis.

4.6 State-augmentation-before-irreducible-history is strong hygiene, not a new theorem

Mori-Zwanzig and related reduced-dynamics work make it clear that unresolved state can generate memory. 33S3 therefore correctly treats state augmentation as a hostile alternative before declaring irreducible hysteresis/history.

Current novelty confidence: low principle; useful integration.

4.7 Recovery-triggered readdressing is the strongest 33S3 residual

The candidate contribution is the conjunction:

track local performance, closure, property-specific law sufficiency and target-relative recoverability separately; test path dependence against missing state; and use the resulting boundary crossings to change the active scale/address without treating degradation, collapse, irreversibility and loss of macro-law sufficiency as synonyms.

Current novelty confidence: moderate candidate synthesis.

5. Generator-level result exposed by the audit

The hostile subtraction changes the interpretation of the family.

The strongest result is **not** that MKUFT independently owns each output statement such as:

- local gain is not whole gain;
- lineage is not functional identity;
- prediction is not intervention authority;
- collapse is not loss of recoverability.

Most of those have substantial prior art in their home domains.

The stronger residual is that one addressed relational engine repeatedly **generates the need for those separations from a common burden**:

A property may cross a scale/layer boundary only when the relation, address and evidence actually preserve the property required by the downstream claim.

This generator then yields operational questions:

- What is the object at this address?
- Which relation is load-bearing for the claimed identity/function?
- Does that relation close or merely correlate?
- Which property descends to the higher address?
- Does the descent survive intervention?
- Has the law failed while the object remains?
- Has closure failed while recovery remains reachable?
- Is apparent history genuine or omitted state?
- Which address is now sufficient for the next propagation step?

In the literature examined for this audit, no single source was located that supplies this full generator as one recursive cross-domain operational architecture.

Current classification: candidate original formal synthesis / operational meta-principle, not yet a new law of nature.

6. Why this is more than a vocabulary synthesis

A weak synthesis merely renames familiar concepts. The 33S2/33S3 family is stronger only if it changes what can be tested.

It does so by generating distinct measurable objects and possible boundary orderings:

- $M_D(q)$ — property-specific higher-law sufficiency margin;
- M_C — constitutive closure margin;
- Rec_H — target- and control-relative recoverability;
- J_l — lower-address performance variable;
- cross-scale sign/alignment classes;
- explicit descent-strength classes;
- prospective readdressing conditions;
- separate predictive and interventional burdens.

This makes it possible to draw and falsify multi-boundary state diagrams rather than treating “emergence”, “health”, “coherence”, “collapse” or “recovery” as one undifferentiated scalar.

That is the practical scientific value even if future literature reduces the historical novelty claim.

7. What this audit does not license

This audit does not license the claims that:

- MKUFT invented emergence, coarse-graining, causal abstraction, lumpability, multilevel selection, resilience, reachability or hysteresis;
- 33S2 is already a new physical law;
- 33S3 is a universal theory of cancer, ecology or recovery;
- similar graph topology proves a common microscopic mechanism across domains;
- the absence of an equivalent source in this targeted search proves global historical priority;
- mathematical/formal novelty proves empirical truth.

8. Promotion path

The strongest residual would earn promotion beyond “candidate synthesis” if held-out applications show that the architecture predicts, before outcome inspection:

1. which variables must be separated by scale/address;
2. which property will lose higher-level sufficiency first;
3. when a lower, higher or coupled-multiscale description will be required;
4. which boundary crossings are distinct and in what order they occur under declared conditions;
5. when adding bounded state/history will restore closure;
6. when a system remains recoverable despite degradation or closure loss;
7. when the same structural rule transfers across domains while mechanism-specific predictions remain correct.

Repeated prospective success of that kind would support calling the residual a **cross-domain invariant principle**. A new physical law would require a further step: a quantitative empirical relation not already entailed by established theory that survives independent experimental tests.

9. Novelty matrix after hostile subtraction

Component	Strongest nearby prior art	Residual status after audit	Current novelty confidence
Micro→macro mapping	causal abstraction, coarse-graining	inherited ingredient	Low
Representative-consistent macro dynamics	lumpability / quotient dynamics	inherited mathematical burden	Low
Property-specific reduced adequacy	goal-oriented/property-driven reduction	inherited principle; new role in law-ownership gate	Low principle / Moderate integration
Predictive vs interventional adequacy	causal abstraction / causal consistency	inherited principle	Low
Functional object ≠ autonomous law for every property	emergence, autonomy, effective dynamics adjacent	explicit separation plus operational gate	Moderate candidate
Lineage → function → closure → law sufficiency	pieces distributed across fields	exact four-object decision chain not located	Moderate candidate
Closure margin M_C vs descent margin $M_D(q)$	adjacent stability/reduction concepts	paired margin inside addressed-law ownership	Moderate candidate
Bidirectional evidence-triggered readdressing	adaptive multiscale modeling broadly	tied specifically to property-law sufficiency without deleting higher object	Moderate candidate
Address before law ownership	effective theories/coarse-graining adjacent	exact recursive rule not located	Moderate-high candidate synthesis
Local vs whole performance	multilevel selection / cancer	inherited principle	Low

Recoverability as constrained reachability	viability/control theory	inherited mathematical family	Low
Hysteresis/asymmetric return	dynamical systems/ecology	inherited principle	Low
State augmentation before irreducible history	Mori–Zwanzig / higher-order closure	inherited principle; useful guard	Low
M_D(q) vs M_C vs Rec_H as separate boundaries	pieces exist separately	exact integrated three-boundary object not located	Moderate candidate
Cross-scale alignment tuple + readdressing	multilevel fitness + addressed-law architecture	integrated operational bridge	Moderate candidate
Full 33S2/33S3 generator	no single equivalent located in searched literature	candidate original formal synthesis / meta-principle	Moderate–high candidate

Confidence concerns novelty relative to the literature examined, not truth, usefulness, patentability or global priority.

10. Current verdict

The hostile audit **does not reduce 33S2 and 33S3 to rediscovery**.

It does substantially narrow the claim. Most memorable one-line outputs are not individually new: science already knows many ways that local and global fitness diverge, that macro models may preserve selected quantities, that interventions are a harder burden than passive prediction, that coarse dynamics can require memory, and that degradation, tipping, hysteresis and recoverability are separable.

What survives is more structurally important:

33S2/33S3 form a candidate original addressed-law decision architecture that separates what makes a higher-order object real from what makes a higher-order law sufficient; assigns that sufficiency property by property; requires interventional burden where causal authority is claimed; permits evidence-driven upward, downward and multiscale readdressing without equating law failure with object failure; and carries the same discipline into local/whole performance, closure, hysteresis and recoverability.

The strongest concise candidate principle is:

Address before law ownership: a higher-order description may own a law only for the property and regime for which descent is demonstrated, and it must surrender or share that ownership when the demonstrated sufficiency fails.

The strongest generator-level conclusion is:

The novelty may reside less in the individual distinctions than in the formal engine that repeatedly derives, types and tests which distinctions are required before information, identity, performance or law authority may cross a scale boundary.

That is significant enough to preserve canonically and test prospectively. It remains open to reduction by stronger prior art or failed empirical use.

11. Primary and close literature used in this audit

- Rubenstein, P. K., Weichwald, S., Bongers, S., Mooij, J. M., Janzing, D., Grosse-Wentrup, M. & Schölkopf, B. (2017). *Causal Consistency of Structural Equation Models*. UAI 2017; arXiv:1707.00819.
- Beckers, S. & Halpern, J. Y. (2019). *Abstracting Causal Models*. AAAI 33, 2678–2685. DOI [10.1609/aaai.v33i01.33012678](https://doi.org/10.1609/aaai.v33i01.33012678); arXiv:1812.03789.
- Beckers, S., Eberhardt, F. & Halpern, J. Y. (2020). *Approximate Causal Abstractions*. PMLR 115, 606–615; arXiv:1906.11583.

- Hoel, E. P., Albantakis, L. & Tononi, G. (2013). *Quantifying causal emergence shows that macro can beat micro*. PNAS 110, 19790–19795. DOI [10.1073/pnas.1314922110](https://doi.org/10.1073/pnas.1314922110).
- Ganguly, A., Petrov, T. & Koeppl, H. (2013). *Markov chain aggregation and its applications to combinatorial reaction networks*. arXiv:1303.4532.
- Geiger, B. C. & Temmel, C. (2012). *Lumpings of Markov chains, entropy rate preservation, and higher-order lumpability*. arXiv:1212.4375.
- Hilder, B. & Sharma, U. (2022). *Quantitative coarse-graining of Markov chains*. arXiv:2201.10256.
- Michaelides, M., Milios, D., Hillston, J. & Sanguinetti, G. (2016). *Property-driven State-Space Coarsening for Continuous Time Markov Chains*. arXiv:1606.01111.
- Bui-Thanh, T., Willcox, K., Ghattas, O. & van Bloemen Waanders, B. (2007). *Goal-oriented, model-constrained optimization for reduction of large-scale systems*. Journal of Computational Physics 224, 880–896. DOI [10.1016/j.jcp.2006.10.026](https://doi.org/10.1016/j.jcp.2006.10.026).
- Legoll, F. & Lelièvre, T. (2010). *Effective dynamics using conditional expectations*. Nonlinearity 23, 2131–2163. arXiv:0906.4865.
- Li, Z., Bian, X., Li, X. & Karniadakis, G. E. (2015). *Incorporation of memory effects in coarse-grained modeling via the Mori–Zwanzig formalism*. Journal of Chemical Physics 143, 243128. DOI [10.1063/1.4935490](https://doi.org/10.1063/1.4935490).
- Gouasmi, A., Parish, E. J. & Duraisamy, K. (2017). *A priori estimation of memory effects in reduced-order models of nonlinear systems using the Mori–Zwanzig formalism*. Proceedings of the Royal Society A 473, 20170385. DOI [10.1098/rspa.2017.0385](https://doi.org/10.1098/rspa.2017.0385); arXiv:1611.06277.
- Goreac, D. (2010). *Viability, Invariance and Reachability for Controlled Piecewise Deterministic Markov Processes Associated to Gene Networks*. arXiv:1002.2242.
- Hellmann, F., Schultz, P., Grabow, C., Heitzig, J. & Kurths, J. (2016). *Survivability of Deterministic Dynamical Systems*. Scientific Reports 6, 29654; arXiv:1506.01257.
- Hammerschmidt, K., Rose, C. J., Kerr, B. & Rainey, P. B. (2014). *Life cycles, fitness decoupling and the evolution of multicellularity*. Nature 515, 75–79. DOI [10.1038/nature13884](https://doi.org/10.1038/nature13884).
- Kon, S. et al. (2017). *Cell competition with normal epithelial cells promotes apical extrusion of transformed cells through metabolic changes*. Nature Cell Biology 19, 530–541. DOI [10.1038/ncb3509](https://doi.org/10.1038/ncb3509).
- Colom, B. et al. (2021). *Mutant clones in normal epithelium outcompete and eliminate emerging tumours*. Nature 598, 510–514. DOI [10.1038/s41586-021-03965-7](https://doi.org/10.1038/s41586-021-03965-7).
- Rindi, L., Dal Bello, M., Dai, L., Gore, J. & Benedetti-Cecchi, L. (2017). *Direct observation of increasing recovery length before collapse of a marine benthic ecosystem*. Nature Ecology & Evolution 1, 0153. DOI [10.1038/s41559-017-0153](https://doi.org/10.1038/s41559-017-0153).